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Jebessa B. Mijena, Erkan Nane

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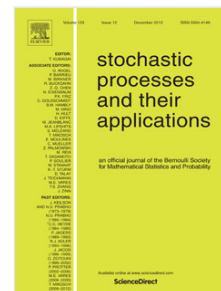
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# SPACE-TIME FRACTIONAL STOCHASTIC PARTIAL DIFFERENTIAL EQUATIONS

JEBESSA B MIJENA AND ERKAN NANE

ABSTRACT. We consider non-linear time-fractional stochastic heat type equation

$$\partial_t^\beta u_t(x) = -\nu(-\Delta)^{\alpha/2} u_t(x) + I_t^{1-\beta}[\sigma(u) \dot{W}(t, x)]$$

in  $(d+1)$  dimensions, where  $\nu > 0$ ,  $\beta \in (0, 1)$ ,  $\alpha \in (0, 2]$  and  $d < \min\{2, \beta^{-1}\}\alpha$ ,  $\partial_t^\beta$  is the Caputo fractional derivative,  $(-\Delta)^{\alpha/2}$  is the generator of an isotropic stable process,  $I_t^{1-\beta}$  is the fractional integral operator,  $\dot{W}(t, x)$  is space-time white noise, and  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is Lipschitz continuous. Time fractional stochastic heat type equations might be used to model phenomenon with random effects with thermal memory. We prove existence and uniqueness of mild solutions to this equation and establish conditions under which the solution is continuous. Our results extend the results in the case of parabolic stochastic partial differential equations obtained in [13; 37]. In sharp contrast to the stochastic partial differential equations studied earlier in [13; 18; 37], in some cases our results give existence of random field solutions in spatial dimensions  $d = 1, 2, 3$ . Under faster than linear growth of  $\sigma$ , we show that time fractional stochastic partial differential equation has no finite energy solution. This extends the result of Foondun and Parshad [14] in the case of parabolic stochastic partial differential equations. We also establish a connection of the time fractional stochastic partial differential equations to higher order parabolic stochastic differential equations.

## 1. INTRODUCTION

Stochastic partial differential equations (SPDE) have been studied in mathematics, and various sciences; see, for example, Khoshnevisan [18] for a big list of references. The area of SPDEs is interesting to mathematicians as it contains big number of hard open problems. SPDE's have been applied in many disciplines that include applied mathematics, statistical mechanics, theoretical physics, theoretical neuroscience, theory of complex chemical reactions, fluid dynamics, hydrology, and mathematical finance. In this paper we introduce new time fractional stochastic partial differential equations (TSPDE) in the sense of Walsh [37], and prove existence and uniqueness of mild solutions. We establish the conditions under which the solution is continuous. When  $\sigma$  grows faster than Lipschitz we show that no finite energy solution exists.

The study of the time-fractional diffusion equations has recently attracted a lot of attention and a typical form of the time fractional equations is  $\partial_t^\beta u = \Delta u$  where  $\partial_t^\beta$  is the Caputo fractional derivative with  $\beta \in (0, 1)$  and  $\Delta = \sum_{i=1}^d \partial_{x_i}^2$  is the Laplacian. These equations are related with anomalous diffusions or diffusions in

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non-homogeneous media, with random fractal structures; see, for instance, [24]. The Caputo fractional derivative  $\partial_t^\beta$  first appeared in [7] is defined for  $0 < \beta < 1$  by

$$\partial_t^\beta u_t(x) = \frac{1}{\Gamma(1-\beta)} \int_0^t \partial_r u_r(x) \frac{dr}{(t-r)^\beta}. \quad (1.1)$$

Its Laplace transform

$$\int_0^\infty e^{-st} \partial_t^\beta u_t(x) dt = s^\beta \tilde{u}_s(x) - s^{\beta-1} u_0(x), \quad (1.2)$$

where  $\tilde{u}_s(x) = \int_0^\infty e^{-st} u_t(x) dt$  and incorporates the initial value in the same way as the first derivative.

Starting from the works [19; 29; 40; 41] much effort has been made in order to introduce a rigorous mathematical approach; see, for example, [28] for a short survey on these results. The solutions to fractional diffusion equations are strictly related with stable densities. Indeed, the stochastic solutions of time fractional diffusions can be realized through time-change by inverse stable subordinators. A couple of recent works in this field are [23; 31].

It might come natural to add just a noise term to the time fractional diffusion and study the equation

$$\partial_t^\beta u_t(x) = \Delta u_t(x) + \dot{W}(t, x); \quad u_t(x)|_{t=0} = u_0(x), \quad (1.3)$$

where  $\dot{W}(t, x)$  is a space-time white noise with  $x \in \mathbb{R}^d$ .

We will make use of **time fractional Duhamel's principle** [34; 36; 35] to get the correct version of (1.3). Let  $G_t(x)$  be the fundamental solution of the time fractional PDE  $\partial_t^\beta u = \Delta u$ . The solution to the time-fractional PDE with force term  $f(t, x)$

$$\partial_t^\beta u_t(x) = \Delta u_t(x) + f(t, x); \quad u_t(x)|_{t=0} = u_0(x), \quad (1.4)$$

is given by Duhamel's principle, the influence of the external force  $f(t, x)$  to the output can be count as

$$\partial_t^\beta V(r, t, x) = \Delta V(r, t, x); \quad V(r, r, x) = \partial_t^{1-\beta} f(t, x)|_{t=r}, \quad (1.5)$$

which has solution

$$V(t, r, x) = \int_{\mathbb{R}^d} G_{t-r}(x-y) \partial_r^{1-\beta} f(r, y) dy.$$

Hence solution to (1.4) is given by

$$u(t, x) = \int_{\mathbb{R}^d} G_t(x-y) u_0(y) dy + \int_0^t \int_{\mathbb{R}^d} G_{t-r}(x-y) \partial_r^{1-\beta} f(r, y) dy dr.$$

Hence if we use time fractional Duhamel's principle we will get the mild (integral) solution of (1.3) to be of the form (informally):

$$\begin{aligned} u(t, x) &= \int_{\mathbb{R}^d} G_t(x-y) u_0(y) dy \\ &\quad + \int_0^t \int_{\mathbb{R}^d} G_{t-r}(x-y) \partial_r^{1-\beta} [\dot{W}(r, y)] dy dr. \end{aligned} \quad (1.6)$$

It is not clear what the fractional derivative of the space-time white noise mean.

We can remove the fractional derivative of the noise term in (1.6) in the following way. Let  $\gamma > 0$ , define the fractional integral by

$$I_t^\gamma f(t) := \frac{1}{\Gamma(\gamma)} \int_0^t (t - \tau)^{\gamma-1} f(\tau) d\tau.$$

For every  $\beta \in (0, 1)$ , and  $g \in L^\infty(\mathbb{R}_+)$  or  $g \in C(\mathbb{R}_+)$

$$\partial_t^\beta I_t^\beta g(t) = g(t).$$

We consider the time fractional PDE with a force given by  $f(t, x) = I_t^{1-\beta} g(t, x)$ , then by the Duhamel's principle the mild solution to (1.4) will be given by

$$u_t(x) = \int_{\mathbb{R}^d} G_t(x - y) u_0(y) dy + \int_0^t \int_{\mathbb{R}^d} G_{t-r}(x - y) g(r, y) dy dr.$$

Hence, the preceding discussion suggest that the correct TFSPDE is the following model problem:

$$\partial_t^\beta u_t(x) = \Delta u_t(x) + I_t^{1-\beta} [\dot{W}(t, x)]; \quad u_t(x)|_{t=0} = u_0(x). \quad (1.7)$$

The fractional integral above in equation (1.7) for functions  $\phi \in L^2(\mathbb{R}^d)$  is defined as

$$\int_{\mathbb{R}^d} \phi(x) I_t^{1-\beta} [\dot{W}(t, x)] dx = \frac{1}{\Gamma(1-\beta)} \int_{\mathbb{R}^d} \int_0^t (t - \tau)^{-\beta} \phi(x) W(d\tau, dx),$$

is well defined only when  $0 < \beta < 1/2$ ! See Walsh isometry (2.13) for this observation.

By the Duhamel's principle, mentioned above, mild (integral) solution of (1.7) will be (informally):

$$u_t(x) = \int_{\mathbb{R}^d} G_t(x - y) u_0(y) dy + \int_0^t \int_{\mathbb{R}^d} G_{t-r}(x - y) W(dy dr). \quad (1.8)$$

Next we want to give a **Physical motivation** to study time fractional SPDEs which is adapted from [9]. The time-fractional SPDEs studied in this paper may arise naturally by considering the heat equation in a material with thermal memory. Let  $u_t(x)$ ,  $k(t, x)$  and  $\vec{H}(t, x)$  denote the body temperature, internal energy and flux density, respectively. Then the relations

$$\begin{aligned} \partial_t k(t, x) &= -\operatorname{div} \vec{H}(t, x), \\ k(t, x) &= \beta u_t(x), \quad \vec{H}(t, x) = -\lambda \nabla u_t(x), \end{aligned} \quad (1.9)$$

yield the classical heat equation  $\beta \partial_t u = \lambda \Delta u$ .

The speed of heat flow is infinite according to the law of classical heat equation, but since the heat flow can be disrupted by the response of the material the propagation speed can be finite. In a material with thermal memory Lunardi and Sinestrari [20], von Wolfersdorf [38] showed that

$$k(t, x) = \bar{\beta} u_t(x) + \int_0^t n(t - s) u_s(x) ds,$$

holds with some appropriate constant  $\bar{\beta}$  and kernel  $n$ . In most cases we would have  $n(t) = \Gamma(1 - \beta_1)^{-1} t^{-\beta_1}$  for  $\beta_1 \in (0, 1)$ . The convolution gives the fact that the

nearer past affects the present more. If in addition the internal energy also depends on past random effects, then

$$\begin{aligned} k(t, x) = & \bar{\beta}u_t(x) + \int_0^t n(t-s)u_s(x)ds \\ & + \int_0^t l(t-s)h(s, u_s(x))W(ds, x), \end{aligned} \quad (1.10)$$

where  $W$  is “white noise” modeling the random effects. Take  $l(t) = \Gamma(2-\beta_2)^{-1}t^{1-\beta_2}$  for  $\beta_2 \in (0, 1)$ , then after differentiation (1.10) gives

$$\partial_t^{\beta_1} u = \operatorname{div} \vec{F} + \frac{1}{\Gamma(2-\beta_2)} \frac{\partial}{\partial t} \int_0^t (t-s)^{1-\beta_2} h(s, u_s(x))W(ds, x). \quad (1.11)$$

In this paper we will study existence and uniqueness of mild solutions to this type of stochastic equations and its extensions:

$$\begin{aligned} \partial_t^\beta u_t(x) = & -\nu(-\Delta)^{\alpha/2} u_t(x) + I_t^{1-\beta}[\sigma(u) \dot{W}(t, x)], \quad \nu > 0, t > 0, x \in \mathbb{R}^d; \\ u_t(x)|_{t=0} = & u_0(x), \end{aligned} \quad (1.12)$$

where the initial datum  $u_0$  is measurable and bounded,  $-(\Delta)^{\alpha/2}$  is the fractional Laplacian with  $\alpha \in (0, 2]$ ,  $\dot{W}(t, x)$  is a space-time white noise with  $x \in \mathbb{R}^d$ , and  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is a Lipschitz function.

One might also consider the following equation in the notation of [9] where  $d = 1$ ,  $\alpha = 2$  but  $\beta_1$  and  $\beta_2$  can be different with  $\beta_2 < \beta_1 + 1/2$

$$\begin{aligned} \partial_t^{\beta_1} u_t(x) = & -\nu(-\Delta)^{\alpha/2} u_t(x) + \partial_t^{\beta_2} \int_0^t \sigma(u(s, x)) \dot{W}(ds, x), \quad \nu > 0, t > 0, x \in \mathbb{R}^d; \\ u_t(x)|_{t=0} = & u_0(x). \end{aligned} \quad (1.13)$$

This space-time white noise case is considered in Section 5.2 of [9]. It can be seen easily with the use of Laplace transforms that Equation (1.12) is equivalent to (1.13) when  $\beta_1 = \beta$  and  $\beta_2 = \beta$ .

A physical important reason to take the fractional integral of the noise in equation (1.12): Apply the fractional derivative of order  $1 - \beta$  to both sides of the equation (1.12) to see the forcing function, in the traditional units  $x/t$ .

Let  $G_t(x)$  be the fundamental solution of the fractional heat type equation

$$\partial_t^\beta G_t(x) = -\nu(-\Delta)^{\alpha/2} G_t(x). \quad (1.14)$$

We know that  $G_t(x)$  is the density function of  $X(E_t)$ , where  $X$  is an isotropic  $\alpha$ -stable Lévy process in  $\mathbb{R}^d$  and  $E_t$  is the first passage time of a  $\beta$ -stable subordinator  $D = \{D_r, r \geq 0\}$ , or the inverse stable subordinator of index  $\beta$ : see, for example, Bertoin [6] for properties of these processes, Baeumer and Meerschaert [4] for more on time fractional diffusion equations, and Meerschaert and Scheffler [25] for properties of the inverse stable subordinator  $E_t$ .

Let  $p_{X(s)}(x)$  and  $f_{E_t}(s)$  be the density of  $X(s)$  and  $E_t$ , respectively. Then the Fourier transform of  $p_{X(s)}(x)$  is given by

$$\widehat{p_{X(s)}}(\xi) = e^{-s\nu|\xi|^\alpha}, \quad (1.15)$$

and

$$f_{E_t}(x) = t\beta^{-1}x^{-1-1/\beta}g_\beta(tx^{-1/\beta}), \quad (1.16)$$

where  $g_\beta(\cdot)$  is the density function of  $D_1$ . The function  $g_\beta(u)$  (cf. Meerschaert and Straka [26]) is infinitely differentiable on the entire real line, with  $g_\beta(u) = 0$  for  $u \leq 0$ .

By conditioning, we have

$$G_t(x) = \int_0^\infty p_{X(s)}(x)f_{E_t}(s)ds. \quad (1.17)$$

A related time-fractional SPDE was studied by Karczewska [17], Chen et al. [9], and Baeumer et al [3]. They have proved regularity of the solutions to the time-fractional parabolic type SPDEs using cylindrical Brownian motion in Banach spaces in line with the methods in [12].

In this paper we study the existence and uniqueness of the solution to (1.12) under global Lipschitz conditions on  $\sigma$ , using the white noise approach of [37]: We say that an  $\mathcal{F}_t$ -adapted random field  $\{u_t(x), t \geq 0, x \in \mathbb{R}^d\}$  is a mild solution of (1.12) with initial value  $u_0$  if the following integral equation is fulfilled

$$u_t(x) = \int_{\mathbb{R}^d} u_0(y)G_t(x-y)dy + \int_0^t \int_{\mathbb{R}^d} \sigma(u_r(y))G_{t-r}(x-y)W(dr dy). \quad (1.18)$$

For a comparison of the two approaches to SPDE's see the paper by Dalang and Quer-Sardanyons [11].

We now briefly give an outline of the paper. We adapt the methods of proofs of the results in [18] with many crucial nontrivial changes. We give some preliminary results in section 2. Moment estimates for time increments and spatial increments of the solution is given in Section 3. The main result in this section is Theorem 1. In section 4, we give main results of the paper about existence and uniqueness, and the continuity of the solution to the time fractional SPDEs. The main result is Theorem 2. In Section 5, we show that under faster than linear growth of  $\sigma$  there is no finite energy solution. In section 6, we discuss the equivalence of time-fractional SPDEs to higher order SPDEs. Throughout the paper, we use the letter  $C$  or  $c$  with or without subscripts to denote a constant whose value is not important and may vary from places to places.

## 2. PRELIMINARIES

In this section, we give some preliminary results that will be needed in the remaining sections of the paper.

Applying the Laplace transform with respect to time variable  $t$ , Fourier transform with respect to space variable  $x$ . Laplace-Fourier transform of  $G$  defined in (1.17) is given by

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}^d} e^{-\lambda t + i\xi \cdot x} G_t(x) dx dt &= \int_0^\infty e^{-\lambda t} dt \int_0^\infty f_{E_t}(s) ds \int_{\mathbb{R}^d} e^{i\xi \cdot x} p_{X(s)}(x) dx \\ &= \int_0^\infty e^{-s\nu|\xi|^\alpha} ds \int_0^\infty e^{-\lambda t} f_{E_t}(s) dt \\ &= \frac{\lambda^\beta}{\lambda} \int_0^\infty e^{-s(\nu|\xi|^\alpha + \lambda^\beta)} ds \end{aligned}$$

$$= \frac{\lambda^{\beta-1}}{\lambda^{\beta} + \nu|\xi|^{\alpha}}, \quad (2.1)$$

here we used the fact that the laplace transform  $t \rightarrow \lambda$  of  $f_{E_t}(u)$  is given by  $\lambda^{\beta-1}e^{-u\lambda^{\beta}}$ . Using the convention,  $\sim$  to denote the Laplace transform and  $*$  the Fourier transform we get

$$\tilde{G}_{\lambda}^{*}(\xi) = \frac{\lambda^{\beta-1}}{\lambda^{\beta} + \nu|\xi|^{\alpha}}. \quad (2.2)$$

Inverting the Laplace transform, it yields

$$G_t^{*}(\xi) = E_{\beta}(-\nu|\xi|^{\alpha}t^{\beta}), \quad (2.3)$$

where

$$E_{\beta}(x) = \sum_{k=0}^{\infty} \frac{x^k}{\Gamma(1 + \beta k)}, \quad (2.4)$$

is the Mittag-Leffler function. In order to invert the Fourier transform, we will make use of the integral [16, eq. 12.9]

$$\int_0^{\infty} \cos(ks) E_{\beta,\alpha}(-as^{\mu}) ds = \frac{\pi}{k} H_{3,3}^{2,1} \left[ \frac{k^{\mu}}{a} \left| \begin{matrix} (1,1), (\alpha,\beta), (1,\mu/2) \\ (1,\mu), (1,1), (1,\mu/2) \end{matrix} \right. \right],$$

where  $\mathcal{R}(\alpha) > 0, \beta > 0, k > 0, a > 0, H_{p,q}^{m,n}$  is the H-function given in [22, Definition 1.9.1, p. 55] and the formula

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\xi x} f(\xi) d\xi = \frac{1}{\pi} \int_0^{\infty} f(\xi) \cos(\xi x) d\xi.$$

Then this gives the function as

$$G_t(x) = \frac{1}{|x|} H_{3,3}^{2,1} \left[ \frac{|x|^{\alpha}}{\nu t^{\beta}} \left| \begin{matrix} (1,1), (1,\beta), (1,\alpha/2) \\ (1,\alpha), (1,1), (1,\alpha/2) \end{matrix} \right. \right]. \quad (2.5)$$

Note that for  $\alpha = 2$  using reduction formula for the H-function we have

$$G_t(x) = \frac{1}{|x|} H_{1,1}^{1,0} \left[ \frac{|x|^2}{\nu t^{\beta}} \left| \begin{matrix} (1,\beta) \\ (1,2) \end{matrix} \right. \right] \quad (2.6)$$

Note that for  $\beta = 1$  it reduces to the Gaussian density

$$G_t(x) = \frac{1}{(4\nu\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{4\nu t}\right). \quad (2.7)$$

Recall uniform estimate of Mittag-Leffler function [33, Theorem 4]

$$\frac{1}{1 + \Gamma(1 - \beta)x} \leq E_{\beta}(-x) \leq \frac{1}{1 + \Gamma(1 + \beta)^{-1}x} \quad \text{for } x > 0. \quad (2.8)$$

**Lemma 1.** For  $d < 2\alpha$ ,

$$\int_{\mathbb{R}^d} G_t^2(x) dx = C^* t^{-\beta d/\alpha} \quad (2.9)$$

where  $C^* = \frac{(\nu)^{-d/\alpha} 2\pi^{d/2}}{\alpha \Gamma(\frac{d}{2})} \frac{1}{(2\pi)^d} \int_0^{\infty} z^{d/\alpha-1} (E_{\beta}(-z))^2 dz$ .

*Proof.* Using Plancherel theorem and (2.3), we have

$$\begin{aligned} \int_{\mathbb{R}^d} |G_t(x)|^2 dx &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |G_t^*(\xi)|^2 d\xi = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |E_\beta(-\nu|\xi|^\alpha t^\beta)|^2 d\xi \\ &= \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \frac{1}{(2\pi)^d} \int_0^\infty r^{d-1} (E_\beta(-\nu r^\alpha t^\beta))^2 dr. \end{aligned} \quad (2.10)$$

$$= \frac{(\nu t^\beta)^{-d/\alpha} 2\pi^{d/2}}{\alpha \Gamma(\frac{d}{2})} \frac{1}{(2\pi)^d} \int_0^\infty z^{d/\alpha-1} (E_\beta(-z))^2 dz. \quad (2.11)$$

We used the integration in polar coordinates for radially symmetric function in the last equation above. Now using equation (2.8) we get

$$\begin{aligned} \int_0^\infty \frac{z^{d/\alpha-1}}{(1 + \Gamma(1-\beta)z)^2} dz &\leq \int_0^\infty z^{d/\alpha-1} (E_\beta(-z))^2 dz \\ &\leq \int_0^\infty \frac{z^{d/\alpha-1}}{(1 + \Gamma(1+\beta)^{-1}z)^2} dz \end{aligned} \quad (2.12)$$

Hence  $\int_0^\infty z^{d/\alpha-1} (E_\beta(-z))^2 dz < \infty$  if and only if  $d < 2\alpha$ . In this case the upper bound in equation (2.12) is

$$\int_0^\infty \frac{z^{d/\alpha-1}}{(1 + \Gamma(1+\beta)^{-1}z)^2} dz = \frac{B(d/\alpha, 2 - d/\alpha)}{\Gamma(1+\beta)^{-d/\alpha}},$$

where  $B(d/\alpha, 2 - d/\alpha)$  is a Beta function.  $\square$

**Remark 1.** For  $d < 2\alpha$ ,

$$\frac{B(d/\alpha, 2 - d/\alpha)}{\Gamma(1+\beta)^{-d/\alpha}} z^{-\beta d/\alpha} \leq \int_0^\infty z^{d/\alpha-1} (E_\beta(-z))^2 dz \leq \frac{B(d/\alpha, 2 - d/\alpha)}{\Gamma(1+\beta)^{-d/\alpha}} z^{-\beta d/\alpha}.$$

$\dot{W}(t, x)$  is a space-time white noise with  $x \in \mathbb{R}^d$ , which is assumed to be adapted with respect to a filtered probability space  $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathbb{P})$ , where  $\mathcal{F}$  is complete and the filtration  $\{\mathcal{F}_t, t \geq 0\}$  is right continuous.  $\dot{W}(t, x)$  is generalized processes with covariance given by

$$\mathbb{E} \left[ \dot{W}(t, x) \dot{W}(s, y) \right] = \delta(x - y) \delta(t - s).$$

That is,  $W(f)$  is a random field indexed by functions  $f \in L^2((0, \infty) \times \mathbb{R}^d)$  and for all  $f, g \in L^2((0, \infty) \times \mathbb{R}^d)$ , we have

$$\mathbb{E} \left[ W(f) W(g) \right] = \int_0^\infty \int_{\mathbb{R}^d} f(t, x) g(t, x) dx dt.$$

Hence  $W(f)$  can be represented as

$$W(f) = \int_0^\infty \int_{\mathbb{R}^d} f(t, x) W(dx dt).$$

Note that  $W(f)$  is  $\mathcal{F}_t$ -measurable whenever  $f$  is supported on  $[0, t] \times \mathbb{R}^d$ .

Next we give the definition of Walsh-Dalang Integrals that is used in equation (1.18). We use the Brownian Filtration  $\{\mathcal{F}_t\}$  and the Walsh-Dalang integrals defined as follows:

- $(t, x) \rightarrow \Phi_t(x)$  is an elementary random field when  $\exists 0 \leq a < b$  and an  $\mathcal{F}_a$ -measurable  $X \in L^2(\Omega)$  and  $\phi \in L^2(\mathbb{R}^d)$  such that

$$\Phi_t(x) = X 1_{[a,b]}(t) \phi(x) \quad (t > 0, x \in \mathbb{R}^d).$$

- If  $h = h_t(x)$  is non-random and  $\Phi$  is elementary, then

$$\int h \Phi dW := X \int_{(a,b) \times \mathbb{R}^d} h_t(x) \phi(x) W(dt dx).$$

- The stochastic integral is Wiener's, and it is well defined iff  $h_t(x) \phi(x) \in L^2([a, b] \times \mathbb{R}^d)$ .
- We have Walsh isometry,

$$\mathbb{E} \left( \left| \int h \Phi dW \right|^2 \right) = \int_0^\infty \int_{\mathbb{R}^d} dy [h_s(y)]^2 \mathbb{E}(|\Phi_s(y)|^2). \quad (2.13)$$

**Remark 2.** One can study SPDEs driven by general time fractional noise that is a generalized process  $\dot{F}(t, x)$  with covariance given by

$$\mathbb{E} \left[ \dot{F}(t, x) \dot{F}(s, y) \right] = \delta(x - y) f(t - s).$$

where  $f(t) = \frac{1}{t^\gamma}$  for  $0 < \gamma < 1$ .

It would be nice to consider relationship between time fractional SPDEs and SPDEs with time fractional noise. We will carry out this project in a forthcoming article.

Let  $\Phi$  be a random field, and for every  $\gamma > 0$  and  $k \in [2, \infty)$  define

$$\mathcal{N}_{\gamma,k}(\Phi) := \sup_{t \geq 0} \sup_{x \in \mathbb{R}^d} (e^{-\gamma t} \|\Phi_t(x)\|_k) = \sup_{t \geq 0} \sup_{x \in \mathbb{R}^d} \left( e^{-\gamma t} \left( \mathbb{E}[\|\Phi_t(x)\|^k] \right)^{1/k} \right). \quad (2.14)$$

If we identify a.s.-equal random fields, then every  $\mathcal{N}_{\gamma,k}$  becomes a norm. Moreover,  $\mathcal{N}_{\gamma,k}$  and  $\mathcal{N}_{\gamma',k}$  are equivalent norms for all  $\gamma, \gamma' > 0$  and  $k \in [2, \infty)$ . Finally, we note that if  $\mathcal{N}_{\gamma,k}(\Phi) < \infty$  for some  $\gamma > 0$  and  $k \in (2, \infty)$ , then  $\mathcal{N}_{\gamma,2}(\Phi) < \infty$  as well, thanks to Jensen's inequality.

**Definition 1.** We denote by  $\mathcal{L}^{\gamma,2}$  the completion of the space of all simple random fields in the norm  $\mathcal{N}_{\gamma,2}$ .

Given a random field  $\Phi := \{\Phi_t(x)\}_{t \geq 0, x \in \mathbb{R}^d}$  and space-time noise  $W$ , we define the [space-time] stochastic convolution  $G \circledast \Phi$  to be the random field that is defined as

$$(G \circledast \Phi)_t(x) := \int_{(0,t) \times \mathbb{R}^d} G_{t-s}(y-x) \Phi_s(y) W(ds dy),$$

for  $t > 0$  and  $x \in \mathbb{R}^d$ , and  $(G \circledast W)_0(x) := 0$ .

We can understand the properties of  $G \circledast \Phi$  for every fixed  $t > 0$  and  $x \in \mathbb{R}^d$  as follows. Define

$$G_s^{(t,x)}(y) := G_{t-s}(y-x) \cdot 1_{(0,t)}(s) \quad \text{for all } s \geq 0 \text{ and } y \in \mathbb{R}^d. \quad (2.15)$$

Clearly,  $G^{(t,x)} \in L^2(\mathbb{R}_+ \times \mathbb{R}^d)$  for  $d < \min\{2, \beta^{-1}\}\alpha$ ; in fact,

$$\int_0^\infty ds \int_{\mathbb{R}^d} [G_s^{(t,x)}(y)]^2 dy = \int_0^t ds \int_{\mathbb{R}^d} [G_s(y)]^2 dy = C^* t^{1-\beta d/\alpha} < \infty.$$

This computation follows from Lemma 1. Thus, we may interpret the random variable  $(G \circledast \Phi)_t(x)$  as the stochastic integral  $\int G_s^{(t,x)} \Phi dW$ , provided that  $\Phi$  is in  $\mathcal{L}^{\beta,2}$  for some  $\beta > 0$ . Let us recall that  $\Phi \mapsto G \circledast \Phi$  is a random linear map; that is, if  $\Phi, \Psi \in \mathcal{L}^{\beta,2}$  for some  $\beta > 0$ , then for all  $a, b \in \mathbb{R}$  the following holds almost surely:

$$\begin{aligned} & \int_{(0,t) \times \mathbb{R}^d} G_{t-s}(y-x)(a\Phi_s(y) + b\Psi_s(y))W(dsdy) \\ &= a \int_{(0,t) \times \mathbb{R}^d} G_{t-s}(y-x)\Phi_s(y)W(dsdy) + b \int_{(0,t) \times \mathbb{R}^d} G_{t-s}(y-x)\Psi_s(y)W(dsdy) \end{aligned}$$

### 3. ESTIMATES ON THE MOMENTS OF THE INCREMENTS OF THE SOLUTION

In this section we prove continuity of various increments related to the solution of (1.12). We start with the next result

**Lemma 2.** *Suppose  $d < \min(2, \beta^{-1})\alpha$ , and  $p \geq 2$ , then we have the following estimates for time increments of the density  $G$ .*

(i). *For  $t \leq t'$ , we have*

$$\int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x-y) - G_{t-r}(x-y)]^2 dr dy \leq \frac{C^*}{(1 - \beta d/\alpha)} (t' - t)^{1 - \beta d/\alpha},$$

where  $C^*$  is a constant given in Lemma 1.

(ii). *For  $x, x' \in \mathbb{R}^d$ , we have*

$$\begin{aligned} c|x' - x|^2 &\leq \int_0^t \int_{\mathbb{R}^d} [G_{t-r}(x-y) - G_{t-r}(x'-y)]^2 dr dy \\ &\leq C|x' - x|^{\min\{(\frac{\alpha - \beta d}{\beta})^-, 2\}}. \end{aligned} \quad (3.1)$$

*Proof.* Using  $G_{t'-r}^*(x - \cdot) - G_{t-r}^*(x - \cdot) = e^{i\xi \cdot x} G_{t'-r}^*(\xi) - e^{i\xi \cdot x} G_{t-r}^*(\xi)$ , Plancherel theorem and computation in Lemma 1 we have

$$\begin{aligned} & \int_{\mathbb{R}^d} |G_{t'-r}(x-y) - G_{t-r}(x-y)|^2 dy \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |G_{t'-r}^*(\xi) - G_{t-r}^*(\xi)|^2 d\xi \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (G_{t'-r}^*(\xi))^2 d\xi + \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (G_{t-r}^*(\xi))^2 d\xi \\ &\quad - \frac{2}{(2\pi)^d} \int_{\mathbb{R}^d} G_{t'-r}^*(\xi) G_{t-r}^*(\xi) d\xi \\ &= C^*(t' - r)^{-\beta d/\alpha} + C^*(t - r)^{-\beta d/\alpha} - \frac{2}{(2\pi)^d} \int_{\mathbb{R}^d} G_{t'-r}^*(\xi) G_{t-r}^*(\xi) d\xi. \end{aligned}$$

Using Plancherel theorem, integration in polar coordinates in  $\mathbb{R}^d$ , and the fact that  $f(z) = E_\beta(-z)$  is decreasing (since it is completely monotonic, i.e.  $(-1)^n f^{(n)}(z) \geq 0$  for all  $z > 0$ ,  $n = 0, 1, 2, 3, \dots$ ), we get

$$\begin{aligned} & \frac{2}{(2\pi)^d} \int_{\mathbb{R}^d} G_{t'-r}^*(\xi) G_{t-r}^*(\xi) d\xi \\ &= \frac{2}{(2\pi)^d} \int_{\mathbb{R}^d} E_\beta(-\nu(t' - r)^\beta |\xi|^\alpha) E_\beta(-\nu(t - r)^\beta |\xi|^\alpha) d\xi \end{aligned}$$

$$\begin{aligned}
 &= \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \frac{2}{(2\pi)^d} \int_0^\infty s^{d-1} E_\beta(-\nu(t'-r)^\beta s^\alpha) E_\beta(-\nu(t-r)^\beta s^\alpha) ds \\
 &\geq \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \frac{2}{(2\pi)^d} \int_0^\infty s^{d-1} E_\beta(-\nu(t'-r)^\beta s^\alpha) E_\beta(-\nu(t'-r)^\beta s^\alpha) ds \\
 &\geq \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \frac{2}{(2\pi)^d} \int_0^\infty s^{d-1} \left( E_\beta(-\nu(t'-r)^\beta s^\alpha) \right)^2 ds \\
 &= \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \frac{2\nu^{-d/\alpha} (t'-r)^{-\beta d/\alpha}}{\alpha(2\pi)^d} \int_0^\infty z^{d/\alpha-1} \left( E_\beta(-z) \right)^2 dz \\
 &= 2C^*(t'-r)^{-\beta d/\alpha}.
 \end{aligned}$$

Now integrating both sides wrt to  $r$  from 0 to  $t$  we get

$$\begin{aligned}
 &\int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x-y) - G_{t-r}(x-y)]^2 dr dy \\
 &\leq \frac{-C^*(t'-t)^{1-\beta d/\alpha}}{1-\beta d/\alpha} + \frac{C^*(t')^{1-\beta d/\alpha}}{1-\beta d/\alpha} + \frac{C^*t^{1-\beta d/\alpha}}{1-\beta d/\alpha} \\
 &+ \frac{2C^*(t'-t)^{1-\beta d/\alpha}}{1-\beta d/\alpha} - \frac{2C^*(t')^{1-\beta d/\alpha}}{1-\beta d/\alpha} \\
 &= \frac{C^*(t'-t)^{1-\beta d/\alpha}}{1-\beta d/\alpha} - \frac{C^*(t')^{1-\beta d/\alpha}}{1-\beta d/\alpha} + \frac{C^*t^{1-\beta d/\alpha}}{1-\beta d/\alpha} \\
 &\leq \frac{C^*(t'-t)^{1-\beta d/\alpha}}{1-\beta d/\alpha},
 \end{aligned} \tag{3.2}$$

the last inequality follows since  $t < t'$ .

Now we prove (ii). Using  $G_{t'-r}^*(x' - \cdot) - G_{t-r}^*(x - \cdot) = e^{i\xi \cdot x'} G_{t'-r}^*(\xi) - e^{i\xi \cdot x} G_{t-r}^*(\xi)$  and Plancherel theorem, we have

$$\begin{aligned}
 &\int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x' - y) - G_{t-r}(x - y)]^2 dy dr \\
 &= \frac{1}{(2\pi)^d} \int_0^t \int_{\mathbb{R}^d} [G_{t'-r}^*(\xi)]^2 \left| 1 - e^{i\xi(x'-x)} \right|^2 d\xi dr
 \end{aligned} \tag{3.3}$$

In the following, we divide our proof into two cases:

Case 1. If  $d+2 < \min(2, \beta^{-1})\alpha$ , then we apply the following inequality

$$\left| 1 - e^{i\xi(x'-x)} \right|^2 = 4 \sin^2 \left( \frac{\langle \xi, x' - x \rangle}{2} \right) \leq |\xi|^2 |x' - x|^2 \tag{3.4}$$

to (3.3) to derive that

$$\begin{aligned}
 &\int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x' - y) - G_{t-r}(x - y)]^2 dr dy \\
 &\leq \frac{1}{(2\pi)^d} \int_0^t \int_{\mathbb{R}^d} [G_{t'-r}^*(\xi)]^2 |\xi|^2 |x' - x|^2 d\xi dr \\
 &= \frac{|x' - x|^2 2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)} \int_0^t \int_0^\infty u^{d+1} (E_\beta(-\nu(t-r)^\beta u^\alpha))^2 du dr \\
 &\leq \frac{|x' - x|^2 2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)} \int_0^t \int_0^\infty \frac{u^{d+1}}{(1 + \Gamma(1 + \beta)^{-1} \nu(t-r)^\beta u^\alpha)^2} dz dr
 \end{aligned}$$

$$= \frac{2\pi^{d/2}B((d+2)/\alpha, 2-(d+2)/\alpha)\nu^{-(d+2)/\alpha}}{(2\pi)^d\Gamma(d/2)\alpha(1-(d+2)\beta/\alpha)\Gamma(1+\beta)^{-(d+2)/\alpha}}t^{1-(d+2)\beta/\alpha}|x'-x|^2.$$

Case 2. If  $d+2 \geq \min(2, \beta^{-1})\alpha$ , then we choose  $\varepsilon > 0$  fixed such that  $d+2\varepsilon < \min(2, \beta^{-1})\alpha$ . [This can be done since  $d < \min(2, \beta^{-1})\alpha$ .] Clearly,  $\varepsilon < 1$ , then we apply the following inequality

$$\left|1 - e^{i\langle \xi, x'-x \rangle}\right|^2 \leq 4^{1-\varepsilon}|\xi|^{2\varepsilon}|x'-x|^{2\varepsilon} \quad (3.5)$$

to (3.3) to derive that

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}^d} [G_{t-r}(x'-y) - G_{t-r}(x-y)]^2 dr dy \\ & \leq \frac{4^{1-\varepsilon}}{(2\pi)^d} \int_0^t \int_{\mathbb{R}^d} [G_{t-r}^*(\xi)]^2 |\xi|^{2\varepsilon} |x'-x|^{2\varepsilon} d\xi dr \\ & = \frac{|x'-x|^{2\varepsilon} 4^{1-\varepsilon} 2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)} \int_0^t \int_0^\infty u^{d+2\varepsilon-1} (E_\beta(-\nu(t-r)^\beta u^\alpha))^2 du dr \\ & \leq \frac{|x'-x|^{2\varepsilon} 4^{1-\varepsilon} 2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)} \int_0^t \int_0^\infty \frac{u^{d+2\varepsilon-1}}{(1+\Gamma(1+\beta)^{-1}\nu(t-r)^\beta u^\alpha)^2} dz dr \\ & = \frac{2\pi^{d/2} 4^{1-\varepsilon} B((d+2\varepsilon)/\alpha, 2-(d+2\varepsilon)/\alpha)\nu^{-(d+2\varepsilon)/\alpha}}{(2\pi)^d \Gamma(d/2)\alpha(1-(d+2\varepsilon)\beta/\alpha)\Gamma(1+\beta)^{-(d+2\varepsilon)/\alpha}} t^{1-(d+2\varepsilon)\beta/\alpha} |x'-x|^{2\varepsilon}. \end{aligned}$$

To estimate lower bound we need to use  $\frac{1}{4}|z| < |e^z - 1| < \frac{7}{4}|z|$  for  $|z| < 1$  [27, Eq. 4.2.38]. Let  $A = \{\xi \in \mathbb{R}_+^d : |\xi| \leq x_0 < \frac{1}{|x'-x|}\}$ .

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}^d} [G_{t-r}(x'-y) - G_{t-r}(x-y)]^2 dy dr \\ & = \frac{1}{(2\pi)^d} \int_0^t \int_{\mathbb{R}^d} [G_{t-r}^*(\xi)]^2 \left|1 - e^{i\xi \cdot (x'-x)}\right|^2 d\xi dr \\ & \geq \frac{1}{4(2\pi)^d} \int_0^t \int_A [G_{t-r}^*(\xi)]^2 |\xi|^2 |x'-x|^2 d\xi dr \\ & = c|x'-x|^2, \end{aligned}$$

this is true because by (2.8)  $E_\beta(-x) < 1$  and using this we have

$$\begin{aligned} \int_0^t \int_A [G_{t-r}^*(\xi)]^2 |\xi|^2 d\xi dr & = \int_0^t \int_A (E_\beta(-\nu|\xi|^\alpha(t-r)^\beta))^2 |\xi|^2 d\xi dr \\ & \leq \int_0^t \int_A |\xi|^2 d\xi dr \\ & \leq x_0^2 t |A|, \end{aligned} \quad (3.6)$$

where  $|A|$  is  $d$ -dimensional volume of  $A$ . This completes the proof.  $\square$

Our results in this section extend the results in [18, Chapter 5] to the time fractional stochastic heat type equations setting. Our presentation in this section follow the presentation in [18, Chapter 5] with many crucial changes. We next prove that the linear map  $\Phi \rightarrow G \otimes \Phi$  is a continuous map from  $\mathcal{L}^{\gamma,2}$  into itself. In particular this will show that if  $\Phi \in \mathcal{L}^{\gamma,2}$ , then  $\Psi_t(x) := (G \otimes \Phi)_t(x)$  is also in  $\mathcal{L}^{\gamma,2}$ , and hence the stochastic convolution  $G \otimes \Phi W$  is also a well-defined random field in  $\mathcal{L}^{\gamma,2}$ .

**Theorem 1.** *If  $\Phi \in \mathcal{L}^{\gamma,2}$  for some  $\gamma > 0$ , then  $G \otimes \Phi$  has a continuous version that is in  $\mathcal{L}^{\gamma,2}$ .*

We will prove this theorem in steps.

The following result extends the analogous result on SPDEs by [10; 13; 21].

**Proposition 1.** *(A stochastic Young inequality). For all  $\gamma > 0, k \in [2, \infty), d < \min\{2, \beta^{-1}\}\alpha$ , and  $\Phi \in \mathcal{L}^{\gamma,2}$ ,*

$$\mathcal{N}_{\gamma,k}(G \otimes \Phi) \leq c_0 k^{1/2} \cdot \mathcal{N}_{\gamma,k}(\Phi)$$

where  $c_0 = \sqrt{4(2\gamma)^{-(1-\beta d/\alpha)}\Gamma(1-\beta d/\alpha)}$

*Proof.* According to the BDG inequality in [18, Proposition 4.4], and Lemma 1, for all  $k \geq 2$ ,

$$\begin{aligned} \|(G \otimes \Phi)_t(x)\|_k^2 &\leq 4k \int_0^t ds \int_{\mathbb{R}^d} dy [G_{t-s}(y-x)]^2 \|\Phi(y)\|_k^2 \\ &\leq 4k [\mathcal{N}_{\gamma,k}(\Phi)]^2 \int_0^t e^{2\gamma s} ds \int_{\mathbb{R}^d} [G_{t-s}(y-x)]^2 dy \\ &= 4k C^* [\mathcal{N}_{\gamma,k}(\Phi)]^2 \int_0^t e^{2\gamma s} (t-s)^{-\beta d/\alpha} ds \\ &= 4k C^* [\mathcal{N}_{\gamma,k}(\Phi)]^2 e^{2\gamma t} \int_0^t e^{-2\gamma u} u^{-\beta d/\alpha} du \\ &\leq 4k C^* [\mathcal{N}_{\gamma,k}(\Phi)]^2 e^{2\gamma t} (2\gamma)^{-(1-\beta d/\alpha)} \Gamma(1-\beta d/\alpha). \end{aligned} \quad (3.7)$$

Divide both sides by  $\exp(2\gamma t)$ , take supremum over all  $(t, x)$ , and then take square roots to finish.  $\square$

**Lemma 3.** *There exists a finite universal constant  $C_1$  such that for all  $\gamma > 0, k \in [1, \infty), t > 0, \Phi \in \mathcal{L}^{\gamma,2}$ ,  $d < \min\{2, \beta^{-1}\}\alpha$  and  $x, x' \in \mathbb{R}^d$ ,*

$$\mathbb{E} \left( |(G \otimes \Phi)_t(x) - (G \otimes \Phi)_t(x')|^k \right) \leq (C_1 k)^{k/2} e^{\gamma k t} [\mathcal{N}_{\gamma,k}(\Phi)]^k \cdot |x' - x|^{\min\{(\frac{\alpha-\beta d}{\beta})^-, 2\} \frac{k}{2}}.$$

*Proof.* We need to only consider when  $\mathcal{N}_{\gamma,k}(\Phi) < \infty$ , a condition that we now assumed. In that case we may apply the BDG inequality to obtain the following

$$\begin{aligned} \|(G \otimes \Phi)_t(x) - (G \otimes \Phi)_t(x')\|_k^2 &= \left\| \int_{(0,t) \times \mathbb{R}^d} [G_{t-s}(y-x) - G_{t-s}(y-x')] \Phi_s(y) W(ds dy) \right\|_k^2 \\ &\leq c_0 k \int_0^t ds \int_{\mathbb{R}^d} dy [G_{t-s}(y-x) - G_{t-s}(y-x')]^2 \|\Phi_s(y)\|_k^2 \end{aligned}$$

Now let us observe that whenever  $0 < s < t$ ,

$$\|\Phi_s(y)\|_k^2 \leq e^{2\gamma s} [\mathcal{N}_{\gamma,k}(\Phi)]^2 \leq e^{2\gamma t} [\mathcal{N}_{\gamma,k}(\Phi)]^2.$$

This and Lemma 2 gives

$$\begin{aligned} \|(G \otimes \Phi)_t(x) - (G \otimes \Phi)_t(x')\|_k^2 &\leq c_0 k e^{2\gamma t} [\mathcal{N}_{\gamma,k}(\Phi)]^2 \int_0^t ds \int_{\mathbb{R}^d} [G_{t-s}(y-x) - G_{t-s}(y-x')]^2 dy \end{aligned}$$

$$\leq Cc_0ke^{2\gamma t}[\mathcal{N}_{\gamma,k}(\Phi)]^2|x' - x|^{\min\{(\frac{\alpha-\beta d}{\beta})^-, 2\}}.$$

Raise both sides to the power of  $k/2$  to finish.  $\square$

**Lemma 4.** *There exists a finite universal constant  $C_1$  such that for every  $\gamma > 0, k \in [1, \infty], t, t' > 0, x \in \mathbb{R}^d, d < \min\{2, \beta^{-1}\}\alpha$  and  $\Phi \in \mathcal{L}^{\gamma,2}$ , then*

$$\mathbb{E}\left(|(G \otimes \Phi)_t(x) - (G \otimes \Phi)_{t'}(x)|^k\right) \leq (C_1k)^{k/2}e^{\gamma kt}[\mathcal{N}_{\gamma,k}(\Phi)]^k \cdot |t - t'|^{(1-\beta d/\alpha)k/2}.$$

*Proof.* Without loss of generality, we suppose that  $0 < t < t'$  and  $\mathcal{N}_{\gamma,k}(\Phi) < \infty$ . In that case, we may write

$$(G \otimes \Phi)_t(x) - (G \otimes \Phi)_{t'}(x) = J_1 + J_2,$$

where

$$J_1 := \int_{(0,t) \times \mathbb{R}^d} [G_{t-s}(y-x) - G_{t'-s}(y-x)]\Phi_s(y)W(dsdy),$$

and

$$J_2 := \int_{(t,t') \times \mathbb{R}^d} G_{t'-s}(y-x)\Phi_s(y)W(dsdy).$$

We apply the BDG inequality and Lemma 1 and 2 to obtain the bound

$$\begin{aligned} \|J_1\|_k^2 &\leq c_0k \int_0^t ds \int_{\mathbb{R}^d} dy [G_{t-s}(y-x) - G_{t'-s}(y-x)]^2 \|\Phi_s(y)\|_k^2 \\ &\leq Cke^{2\gamma t}[\mathcal{N}_{\gamma,k}(\Phi)]^2(t'-t)^{1-\beta d/\alpha} \end{aligned}$$

and

$$\begin{aligned} \|J_2\|_k^2 &\leq c_0k \int_t^{t'} ds \int_{\mathbb{R}^d} dy [G_{t'-s}(y-x)]^2 \|\Phi_s(y)\|_k^2 \\ &\leq Cke^{2\gamma t'}[\mathcal{N}_{\gamma,k}(\Phi)]^2 \int_t^{t'} (t'-s)^{-\beta d/\alpha} ds \\ &\leq Cke^{2\gamma t'}[\mathcal{N}_{\gamma,k}(\Phi)]^2(t'-t)^{1-\beta d/\alpha}. \end{aligned}$$

In this way we obtain the bound

$$\begin{aligned} \|(G \otimes \Phi)_t(x) - (G \otimes \Phi)_{t'}(x)\|_k &\leq 2\|J_1\|_k^2 + 2\|J_2\|_k^2 \\ &\leq Cke^{2\gamma t}[\mathcal{N}_{\gamma,k}(\Phi)]^2(t'-t)^{1-\beta d/\alpha}, \end{aligned}$$

where  $C$  is a finite constant. Hence, we deduce the lemma after we raise both sides of the preceding display to the power  $k/2$ .  $\square$

In the following, we consider

$$U_t(x) = \int_0^t \int_{\mathbb{R}^d} G_{t-r}(x-y)W(dr dy). \quad (3.8)$$

This is the random part of the mild solution to (1.12) when  $\sigma \equiv 1$ .

**Proposition 2.** *Suppose  $d < \min(2, \beta^{-1})\alpha$ , and  $k \geq 2$ , then we have the following moment estimates for time increments and spatial increments, respectively.*

(i). *For  $t \leq t'$ , we have*

$$c_9^{-1}|t' - t|^{(1-\frac{\beta d}{\alpha})\frac{k}{2}} \leq \mathbb{E}[|U_{t'}(x) - U_t(x)|^k] \leq c_9|t' - t|^{(1-\frac{\beta d}{\alpha})\frac{k}{2}}. \quad (3.9)$$

(ii). For  $x, x' \in \mathbb{R}^d$ , we have

$$c|x' - x|^k \leq \mathbb{E}[|U_t(x) - U_t(x')|^k] \leq c_{10}|x - x'|^{\min\{(\frac{\alpha-\beta d}{\beta})^-, 2\}\frac{k}{2}}. \quad (3.10)$$

*Proof.* We prove the lower bound in (i) at first. By the Hölder's inequality, the fact that the stochastic integrals over nonoverlapping time intervals are independent, isometry of the stochastic integrals and the proof of Lemma 2, we have that for  $k \geq 2$ ,  $t \leq t'$

$$\begin{aligned} \mathbb{E}[|U_{t'}(x) - U_t(x)|^k] &\geq (\mathbb{E}[|U_{t'}(x) - U_t(x)|^2])^{k/2} \\ &= \left( \mathbb{E} \left| \int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x-y) - G_{t-r}(x-y)] W(dydr) + \int_t^{t'} \int_{\mathbb{R}^d} G_{t'-r}(x-y) W(dydr) \right|^2 \right)^{k/2} \\ &= \left( \int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x-y) - G_{t-r}(x-y)]^2 dydr + \int_t^{t'} \int_{\mathbb{R}^d} G_{t'-r}^2(x-y) dydr \right)^{k/2} \\ &\geq \left( \frac{C^*}{1 - \beta d/\alpha} \right)^{k/2} |t' - t|^{(1 - \frac{\beta d}{\alpha})\frac{k}{2}}. \end{aligned} \quad (3.11)$$

Now, for the upper bound, applying the Burkholder's inequality, the fact that  $|a + b|^k \leq 2^k(|a|^k + |b|^k)$ , and Lemma 2 we have that

$$\begin{aligned} &\mathbb{E}[|U_{t'}(x) - U_t(x)|^k] \\ &\leq c_2 \left( \int_0^t \int_{\mathbb{R}^d} [G_{t'-r}(x-y) - G_{t-r}(x-y)]^2 dydr \right)^{k/2} \\ &\quad + c_2 \left( \int_t^{t'} \int_{\mathbb{R}^d} G_{t'-r}^2(x-y) dydr \right)^{k/2} \\ &\leq c_2 \left( \frac{C^*(t' - t)^{1 - \beta d/\alpha}}{1 - \beta d/\alpha} \right)^{k/2} + c_2 \left( \frac{C^*(t' - t)^{1 - \beta d/\alpha}}{1 - \beta d/\alpha} \right)^{k/2} \\ &= 2c_2 \left( \frac{C^*}{1 - \beta d/\alpha} \right)^{k/2} |t' - t|^{(1 - \frac{\beta d}{\alpha})\frac{k}{2}}. \end{aligned}$$

We now prove (ii). For  $p \geq 2$  with  $x, x' \in \mathbb{R}^d$ , by the Burkholder's inequality and Lemma 2, we have

$$\begin{aligned} \mathbb{E}[|U_t(x') - U_t(x)|^k] &\leq c_3 \left( \int_0^t \int_{\mathbb{R}^d} [G_{t-r}(x' - y) - G_{t-r}(x - y)]^2 dydr \right)^{k/2} \\ &= C|x - x'|^{\min\{(\frac{\alpha-\beta d}{\beta})^-, 2\}\frac{k}{2}}. \end{aligned}$$

To prove the lower bound we use the fact that  $k \geq 2$  and Lemma 2(ii)

$$\begin{aligned} \mathbb{E}[|U_t(x) - U_t(x')|^k] &\geq (\mathbb{E}[|U_t(x) - U_t(x')|^2])^{k/2} \\ &= \left( \int_0^t \int_{\mathbb{R}^d} [G_{t-r}(x' - y) - G_{t-r}(x - y)]^2 dydr \right)^{k/2} \\ &\geq c|x' - x|^k. \end{aligned}$$

□

**PROOF OF THEOREM 1.** If  $\Phi \in \mathcal{L}^{\gamma,2}$  for some  $\gamma > 0$ , then  $G \circledast \Phi$  is an adapted random field. This can be seen by considering simple random fields  $\Phi$ , as limits of adapted random fields are adapted. But the result is easy to deduce when  $\Phi$  is simple.

Lemma 3 and Lemma 4 shows that  $G \circledast \Phi$  is continuous in  $L^2(\Omega)$ , then Proposition 4.6 in [18] implies that  $G \circledast \Phi$  is in  $\mathcal{L}^{\gamma,2}$ . We give some details of the preceding.

Let us choose and fix  $t > 0$  and  $x \in \mathbb{R}^d$ . We need to show that  $(\tau, z) \rightarrow (G^{(t,x)} \circledast \Phi)_\tau(z)$  is continuous in  $L^2(\Omega)$  to show that  $G \circledast \Phi$  is in  $\mathcal{L}^{\gamma,2}$ .

Since

$$\begin{aligned} & \mathbb{E} \left( \left| (G^{(t,x)} \circledast \Phi)_\tau(z) - (G^{(t,x)} \circledast \Phi)_\tau(z') \right|^2 \right) \\ &= \mathbb{E} \left( \left| (G^{(t,0)} \circledast \Phi)_\tau(z+x) - (G^{(t,0)} \circledast \Phi)_\tau(z'+x) \right|^2 \right), \end{aligned} \quad (3.12)$$

the proof of Lemma 3 ensures that the preceding tends to zero, uniformly on  $(\tau, z, z') \in (0, t) \times (-n, n)^d \times (-n, n)^d$  for any  $n > 0$  fixed, as  $|z - z'| \rightarrow 0$ . This shows uniform continuity in  $L^2(\Omega)$ , in the space variable.

On the other hand, we may follow the steps in the proof of Lemma 4 to deduce that

$$\begin{aligned} & \mathbb{E} \left( \left| (G^{(t,x)} \circledast \Phi)_\tau(z) - (G^{(t,x)} \circledast \Phi)_{\tau'}(z) \right|^2 \right) \\ &= \int_0^{\tau'-\tau} dr \int_{\mathbb{R}^d} dy [G_{t-s}(y - z - x)]^2 \mathbb{E} \left( |\Phi_s(y)|^2 \right) \end{aligned} \quad (3.13)$$

for  $0 < \tau < \tau'$ . Then we get

$$\begin{aligned} & \mathbb{E} \left( \left| (G^{(t,x)} \circledast \Phi)_\tau(z) - (G^{(t,x)} \circledast \Phi)_{\tau'}(z) \right|^2 \right) \\ &= [\mathcal{N}_{\gamma,2}(\Phi)]^2 \int_0^{\tau'-\tau} e^{2\gamma s} dr \int_{\mathbb{R}^d} dy [G_{t-s}(y - z - x)]^2. \end{aligned} \quad (3.14)$$

The preceding quantity goes to zero, as  $\tau' - \tau \rightarrow 0$ , uniformly for all  $0 < \tau < \tau' < t$ , and  $z \in \mathbb{R}^d$ . This proves the remaining  $L^2$ -continuity in the time variable, and completes the proof of the fact that  $G \circledast \Phi \in \mathcal{L}^{\gamma,2}$ .

The continuity of a modification of the stochastic convolution  $G \circledast \Phi$  follows from Lemmas 3 and 4 by using a suitable form of the Kolmogorov continuity theorem [18, Theorem C.6]  $\square$

#### 4. EXISTENCE AND UNIQUENESS OF SOLUTIONS TO THE TIME FRACTIONAL SPDES

Recall that  $\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is Lipchitz continuous. This means that there exists  $Lip > 0$  such that

$$|\sigma(x) - \sigma(y)| \leq Lip|x - y| \quad \text{for all } x, y \in \mathbb{R}. \quad (4.1)$$

We may assume, without loss of generality, that  $Lip$  is also greater than  $|\sigma(0)|$ . Since  $|\sigma(x)| \leq |\sigma(0)| + Lip|x|$ , it follows that  $|\sigma(x)| \leq Lip(1 + |x|)$  for all  $x \in \mathbb{R}$ .

The next theorem establishes existence and uniqueness of mild solutions of (1.12). It is our first main result in this paper.

**Theorem 2.** Let  $d < \min\{2, \beta^{-1}\}\alpha$ . If  $\sigma$  is Lipschitz continuous and  $u_0$  is measurable and bounded, then there exists a continuous random field  $u \in \cup_{\gamma>0} \mathcal{L}^{\gamma,2}$  that solves (1.12) with initial function  $u_0$ . Moreover,  $u$  is a.s.-finite among all random fields that satisfy the following: There exists a positive and finite constant  $L$ -depending only on  $Lip$ , and  $\sup_{z \in \mathbb{R}^d} |u_0(z)|$ - such that

$$\sup_{x \in \mathbb{R}^d} \mathbb{E} \left( |u_t(x)|^k \right) \leq L^k \exp(Lk^{1+\alpha/(\alpha-\beta d)}t). \quad (4.2)$$

**Remark 3.** Theorem 2 implies that a random field solution exists when  $d < \min\{2, \beta^{-1}\}\alpha$ . So in the case  $\alpha = 2, \beta < 1/2$ , a random field solution exists when  $d = 1, 2, 3$ . The analogous result in the case  $\beta = 1$  is only valid when  $d = 1$ . The assumption  $d < \min\{2, \beta^{-1}\}\alpha$  is essential for the second moments of the solution to be finite. For example, if  $d \geq 2\alpha$  in the proof of Lemma 1 the left hand side of (2.9) diverges. So, if  $d \geq \min\{2, \beta^{-1}\}\alpha$  the second moments of the solution doesn't exist as a finite number.

**Remark 4.** Is the power  $k^{1+\alpha/(\alpha-\beta d)}$  in Theorem 2 artificial? Is it possible to find a lower bound for the moments with the same power of  $k$ ?

We use Picard iteration to prove Theorem 2 that is outlined in [18, Chapter 1] with crucial changes. Define  $u_t^{(0)}(x) := u_0(x)$ , and iteratively define  $u_t^{(n+1)}$  from  $u_t^{(n)}$  as follows:

$$\begin{aligned} u_t^{(n+1)}(x) &:= (G_t * u_0)(x) + \int_{(0,t) \times \mathbb{R}^d} G_{t-r}(x-y) \sigma(u^{(n)}(r,y)) W(dr dy) \\ &:= (G_t * u_0)(x) + (G \otimes \sigma(u^{(n)}))_t(x) \end{aligned} \quad (4.3)$$

for all  $n \geq 0, t > 0$ , and  $x \in \mathbb{R}^d$ . Moreover, we set  $u_0^{(k)}(x) := u_0(x)$  for every  $k \geq 1$  and  $x \in \mathbb{R}^d$ .

**Proposition 3.** The random fields  $\{u^{(n+1)}\}_{n=0}^\infty$  are well-defined, and each is in  $\cup_{\gamma>0} \mathcal{L}^{\gamma,2}$ . Moreover, there exist positive and finite constants  $L_1$  and  $L_2$ -depending only on  $Lip$ , and  $\sup_{z \in \mathbb{R}^d} |u_0(z)|$ -such that

$$\sup_{x \in \mathbb{R}^d} \mathbb{E} \left( |u_t^{(n)}(x)|^k \right) \leq L_1^k \exp(L_2 k^{1+\alpha/(\alpha-\beta d)}t), \quad (4.4)$$

simultaneously for all  $k \in [1, \infty)$ ,  $n \geq 0$ , and  $t > 0$ .

*Proof.* We prove this proposition using induction on  $n$ . Since  $u_0$  is non-random, bounded, and measurable, it is in  $\cup_{\gamma>0} \mathcal{L}^{\gamma,2}$ . Moreover (4.4) holds since

$$\sup_{x \in \mathbb{R}^d} \mathbb{E} \left( |u_t^{(0)}(x)|^k \right) = \sup_{z \in \mathbb{R}^d} |u_0(z)| \leq L_1^k \exp(L_2 k^{1+\alpha/(\alpha-\beta d)}t).$$

Next suppose that the proposition holds for some integer  $n \geq 0$ . We will show the proposition for  $n+1$ .

Let  $k$  denote a fixed real number  $\geq 2$ . Now

$$u_t^{(l+1)}(x) := (G_t * u_0)(x) + B_t^{(l)}(x), \quad (4.5)$$

where

$$B_t^{(l)}(x) := \int_{(0,t) \times \mathbb{R}^d} G_{t-r}(x-y) \sigma(u^{(l)}(r,y)) W(dr dy),$$

for all  $t > 0, x \in \mathbb{R}^d$ , and  $l \geq 0$ .

It is easy to see that  $|(G_t * u_0)(x)| \leq \sup_{z \in \mathbb{R}^d} |u_0(z)|$ , hence for all  $\gamma > 0$

$$\mathcal{N}_{\gamma,k}((G * u_0)) \leq \sup_{z \in \mathbb{R}^d} |u_0(z)|. \quad (4.6)$$

We estimate  $B^{(n)}$  using the Stochastic Young inequality in Proposition 1:

$$\begin{aligned} \mathcal{N}_{\gamma,k}(B^{(n)}) &\leq c_{\alpha,\beta,d} \frac{\sqrt{k}}{\gamma^{1-\beta d/\alpha}} \mathcal{N}_{\gamma,k}(\sigma(u^{(n)})) \\ &\leq c_{\alpha,\beta,d} \cdot Lip \cdot \frac{\sqrt{k}}{\sqrt{\gamma^{1-\beta d/\alpha}}} \cdot \left\{ 1 + \mathcal{N}_{\gamma,k}(u^{(n)}) \right\}, \end{aligned} \quad (4.7)$$

where  $c_{\alpha,\beta,d}$  is the constant in Proposition 1.

Now combining equations (4.6) and (4.7) we get for all  $\gamma > 0$

$$\begin{aligned} \mathcal{N}_{\gamma,k}(u^{(n+1)}) &\leq \sup_{z \in \mathbb{R}^d} |u_0(z)| + c_{\alpha,\beta,d} \cdot Lip \cdot \frac{\sqrt{k}}{\sqrt{\gamma^{1-\beta d/\alpha}}} \\ &\quad + c_{\alpha,\beta,d} \cdot Lip \cdot \frac{\sqrt{k}}{\sqrt{\gamma^{1-\beta d/\alpha}}} \cdot \mathcal{N}_{\gamma,k}(u^{(n)}). \end{aligned} \quad (4.8)$$

We can find a constant  $L_2 > 0$ —depending only on  $Lip, \alpha, \beta, d$ —such that  $\gamma := L_2 k^{\alpha/(\alpha-\beta d)}$  satisfies

$$c_{\alpha,\beta,d} \cdot Lip \cdot \frac{\sqrt{k}}{\sqrt{\gamma^{1-\beta d/\alpha}}} \leq \frac{1}{4},$$

then

$$\mathcal{N}_{\gamma,k}(u^{(n+1)}) \leq \sup_{z \in \mathbb{R}^d} |u_0(z)| + \frac{1}{4} + \frac{1}{4} \mathcal{N}_{\gamma,k}(u^{(n)}).$$

In order to simplify notation, let

$$\theta = \sup_{z \in \mathbb{R}^d} |u_0(z)| + \frac{1}{4}.$$

We solve recursively to find that

$$\begin{aligned} \mathcal{N}_{\gamma,k}(u^{(n+1)}) &\leq \theta + \frac{\theta}{4} + \frac{1}{4^2} \mathcal{N}_{\gamma,k}(u^{(n-1)}) \\ &\leq \dots \leq \sum_{j=1}^n \frac{\theta}{4^j} + \frac{1}{4^{n+1}} \mathcal{N}_{\gamma,k}(u^{(0)}). \end{aligned}$$

But

$$\mathcal{N}_{\gamma,k}(u^{(0)}) = \sup_{t \geq 0} \sup_{x \in \mathbb{R}^d} (e^{-\gamma t} |u_0(x)|) = \sup_{z \in \mathbb{R}^d} |u_0(z)|.$$

Thus, for our choice of  $\gamma$ , we have

$$\mathcal{N}_{\gamma,k}(u^{(n+1)}) \leq \frac{4}{3} \theta + 4^{-(n+1)} \sup_{z \in \mathbb{R}^d} |u_0(z)| \leq \frac{4}{3} \theta + \sup_{z \in \mathbb{R}^d} |u_0(z)| = L_1.$$

Therefore by Theorem 1 and induction we get  $u^{(n+1)} \in \cup_{\gamma>0} \mathcal{L}^{\gamma,2}$ . Moreover, the last inequality means:

$$\mathbb{E} \left( \left| u_t^{(n+1)}(x) \right|^k \right) L_1^k e^{\gamma k t} = L_1^k \exp(L_2 k^{1+\alpha/(\alpha-\beta d)} t).$$

Hence, (4.4) holds for  $n + 1$ , and this proves the induction step, and proves the proposition.  $\square$

**PROOF OF THEOREM 2.** Our proof follows similar steps as in the proof of Theorem 5.5 in [18] with nontrivial crucial changes. Let's choose and fix some  $k \in [2, \infty)$ . Let us write

$$\begin{aligned} J &:= u_t^{(n+1)}(x) - u_t^{(n)}(x) \\ &= \int_{(0,t) \times \mathbb{R}^d} G_{t-r}(x-y) \left[ \sigma(u^{(n)}(r,y)) - \sigma(u^{(n-1)}(r,y)) \right] W(dr dy). \end{aligned}$$

By proposition 3 every  $u^{(n)} \in \mathcal{L}^{\gamma,2}$  for some  $\gamma > 0$ , and hence  $J$  is a well-defined stochastic integral for every  $n \geq 0$ . By the same proposition we can also choose a continuous version of  $(t,x) \rightarrow u_t^{(n)}(x)$  for every  $n$ .

Next we estimate  $J$ . We apply the BDG inequality (Propositon 4.4 in [18]) to bound  $\|J\|_k$  as follows:

$$\begin{aligned} \|J\|_k^2 &\leq C \cdot k \int_0^t dr \int_{\mathbb{R}^d} dy [G_{t-r}(x-y)]^2 \left\| \sigma(u_r^{(n)}(y)) - \sigma(u_r^{(n-1)}(y)) \right\|_k^2 \\ &\leq C \cdot Lip^2 k \int_0^t dr \int_{\mathbb{R}^d} dy [G_{t-r}(x-y)]^2 \left\| u_r^{(n)}(y) - u_r^{(n-1)}(y) \right\|_k^2 \\ &\leq C \cdot Lip^2 k \left[ \mathcal{N}_{\gamma,k} \left( u^{(n)} - u^{(n-1)} \right) \right]^2 \int_0^t e^{2\gamma r} dr \int_{\mathbb{R}^d} dy [G_{t-r}(x-y)]^2 \end{aligned}$$

Since  $\int_{\mathbb{R}^d} dy [G_{t-r}(x-y)]^2 = C^*(t-r)^{-\beta d/\alpha}$ , it follows that

$$\begin{aligned} \|J\|_k^2 &\leq C \cdot Lip^2 k e^{2\gamma t} \left[ \mathcal{N}_{\gamma,k} \left( u^{(n)} - u^{(n-1)} \right) \right]^2 \int_0^t \frac{e^{-2\gamma r}}{r^{\beta d/\alpha}} dr \\ &\leq \frac{C \cdot Lip^2 k}{\gamma^{1-\beta d/\alpha}} e^{2\gamma t} \left[ \mathcal{N}_{\gamma,k} \left( u^{(n)} - u^{(n-1)} \right) \right]^2. \end{aligned}$$

Since the right hand side does not depend on  $(t,x)$  after dividing by  $e^{2\gamma t}$ , we optimize over  $(t,x)$  to find the recursive inequality,

$$\mathcal{N}_{\gamma,k} \left( u^{(n+1)} - u^{(n)} \right) \leq C \cdot Lip \mathcal{N}_{\gamma,k} \left( u^{(n)} - u^{(n-1)} \right) \left[ \frac{\sqrt{k}}{\sqrt{\gamma^{1-\beta d/\alpha}}} \right].$$

The preceding holds for all  $\gamma > 0$ . We can choose  $\gamma_0 := qk^{\alpha/(\alpha-\beta d)}$ , where  $q > L_2$  depends only on  $L_2$  and ensures that

$$\mathcal{N}_{\gamma_0,k} \left( u^{(n+1)} - u^{(n)} \right) \leq \frac{1}{4} \mathcal{N}_{\gamma_0,k} \left( u^{(n)} - u^{(n-1)} \right).$$

According to Proposition 3

$$\mathcal{N}_{\gamma_0,k} \left( u^{(1)} - u^{(0)} \right) \leq \mathcal{N}_{\gamma_0,k} \left( u^{(1)} \right) + \mathcal{N}_{\gamma_0,k} \left( u^{(0)} \right).$$

Since  $q > L_2$ , Proposition 3 implies that

$$\mathcal{N}_{\gamma_0,k} \left( u^{(1)} \right) \vee \mathcal{N}_{\gamma_0,k} \left( u^{(0)} \right) \leq L_1.$$

Hence we obtain the estimate

$$\mathcal{N}_{\gamma_0, k} \left( u^{(n+1)} - u^{(n)} \right) \leq \frac{L_1}{4^n},$$

valid for all  $n \geq 0$  and  $k \in [2, \infty)$ . From this we obtain

- (1) The random field  $u := \lim_{n \rightarrow \infty} u^{(n)}$  exists, where the limit takes place almost surely and in every norm  $\mathcal{N}_{\gamma_0, k}$ ;
- (2) the random field  $\mathbb{S}$  defined by

$$\mathbb{S}_t(x) := \lim_{n \rightarrow \infty} \int_{(0, t) \times \mathbb{R}^d} G_{t-r}(x-y) \sigma(u^{(n)}(r, y)) W(dr dy),$$

exists, where the limit takes place almost surely and in every norm  $\mathcal{N}_{\gamma_0, k}$ .

Since the function  $(G_t * u_0)(x)$  is smooth for  $t > 0$ , combining Lemmas 3 and 4, applied to  $\Phi := u^{(n)}$ , with equation (4.4) we see that for every  $k \in [2, \infty)$  and  $\tau \in (0, \infty)$  there exists a finite constant  $A_{k, \tau}$  such that

$$\mathbb{E} \left( |u_t^{(n)}(x) - u_{t'}^{(n)}(x')|^k \right) \leq A_{k, \tau} \left( |x - x'|^{\min\left\{\left(\frac{\alpha - \beta d}{\beta}\right)^-, 2\right\} \frac{k}{2}} + |t - t'|^{\frac{k(1 - \beta d / \alpha)}{2}} \right),$$

simultaneously for all  $t, t' \in [0, \tau]$ ,  $x, x' \in \mathbb{R}^d$ . The right hand side of this inequality does not depend on  $n$ . Hence, using Fatou's lemma we get

$$\mathbb{E} \left( |u_t(x) - u_{t'}(x')|^k \right) \leq A_{k, \tau} \left( |x - x'|^{\min\left\{\left(\frac{\alpha - \beta d}{\beta}\right)^-, 2\right\} \frac{k}{2}} + |t - t'|^{\frac{k(1 - \beta d / \alpha)}{2}} \right). \quad (4.9)$$

Now by a suitable form of Kolmogorov continuity theorem in [18, Theorem c.6] we get that  $u$  has a version that is continuous. Moreover, (4.2) holds with  $L = \max\{q, L_1\}$  since  $\mathcal{N}_{\gamma_0, k}(u) \leq L_1$  for all  $k \in [2, \infty)$ .

So far, we know that

$$u_t(x) = (G_t * u_0)(x) + \mathbb{S}_t(x), \quad (4.10)$$

where the equality is understood in the sense that the  $\mathcal{N}_{\gamma_0, k}$ -norm of the difference between the two sides of that inequality is zero. Equivalently, by the use of Fubini theorem we have shown that with probability one, the identity (4.10) holds for almost every  $t > 0$  and  $x \in \mathbb{R}^d$ .

Since  $u = \lim_{n \rightarrow \infty} u^{(n)}$  and  $u^{(n)} \in \mathcal{L}^{\gamma, 2}$  for some  $\gamma > 0$  that is independent of  $n$ , we can conclude that  $u \in \mathcal{L}^{\gamma, 2}$ , and hence

$$\bar{\mathbb{S}}_t(x) := \int_{(0, t) \times \mathbb{R}^d} G_{t-r}(x-y) \sigma(u(r, y)) W(dr dy),$$

is well-defined for all  $t > 0$ , and  $x \in \mathbb{R}^d$ . Now we apply the Fatou's Lemma together with Proposition 1 with  $\Phi = u^{(n)} - u$  and  $\gamma_0 = qk^{\alpha/(\alpha - \beta d)}$  to get that

$$\mathcal{N}_{\gamma_0, k}(\mathbb{S} - \bar{\mathbb{S}}) \leq C \cdot \liminf_{n \rightarrow \infty} \mathcal{N}_{\gamma_0, k} \left( u^{(n)} - u \right) = 0.$$

Hence  $\mathbb{S}$  and  $\bar{\mathbb{S}}$  are versions of one another. Another application of Lemmas 3 and 4 shows that  $\mathbb{S}$  has a continuous version. We combine these results with (4.10) to see that the present version of  $u$  is a mild solution—in the sense of (1.18) for the right versions of the integrals—for the time fractional stochastic heat type equation (1.12). This completes the proof of existence.

Suppose  $v \in \mathcal{L}^{\gamma, 2}$  is another random field that is mild solution to (1.12) with initial function  $u_0$ . We can argue similar to the arguments above to show that if  $Q$

is sufficiently large,  $\gamma_1 = Qk^{\alpha/(\alpha-\beta d)}$ , then  $\mathcal{N}_{\gamma_1,k}(u-v) \leq \frac{1}{4}\mathcal{N}_{\gamma_1,k}(u-v)$ . Hence  $u$  and  $v$  are versions of one another. The other details are omitted.  $\square$

**Corollary 1.** *Let  $u(t, x)$  be a solution of (1.12). For any  $\eta < \frac{1}{2} \min \left\{ \left( \frac{\alpha-\beta d}{\beta} \right)^-, 2 \right\}$ , the solution  $u(t, x)$  has a version that has Hölder continuous paths with exponent  $\eta$  as a function  $x \rightarrow u_t(x)$  for any fixed time  $t$ . For any  $\nu < \frac{(1-\beta d/\alpha)}{2}$ , the solution  $u(t, x)$  has a version that has Hölder continuous paths with exponent  $\nu$  as a function  $t \rightarrow u_t(x)$  for any fixed  $x$ .*

*Proof.* The proof follows by (4.9) and an application of the Kolmogorov's criterion for Hölder continuity of stochastic processes: see, for example, Theorem I.2.1 in Revuz and Yor [32].  $\square$

## 5. NON-EXISTENCE OF SOLUTIONS

In this section, we will establish the non-existence of finite energy solutions when  $\sigma$  grows faster than linear. We say that a random field  $u$  is a finite energy solution to the stochastic heat equation (1.12) when  $u \in \cup_{\gamma>0} \mathcal{L}^{\gamma,2}$  and there exists  $\rho_* > 0$  such that

$$\int_0^\infty e^{-\rho_* t} \mathbb{E}(|u_t(x)|^2) dt < \infty \quad \text{for all } x \in \mathbb{R}^d.$$

**Remark 5.** *If  $\rho \in (0, \infty)$ , then*

$$\int_0^\infty e^{-\rho t} \mathbb{E}(|u_t(x)|^2) dt \leq [\mathcal{N}_{\gamma,2}(u)]^2 \cdot \int_0^\infty e^{-(\rho-2\gamma)t} dt.$$

*Therefore if  $\rho > 2\gamma$  and  $\mathcal{N}_{\gamma,2}(u) < \infty$ , then the preceding integral is finite. By Theorem 2, when  $\sigma$  is Lipschitz-continuous function and  $u_0$  is bounded and measurable, then there exists a finite energy solution to the time fractional stochastic type equation (1.12).*

When  $\sigma$  is Lipschitz continuous then it is at most linear growth. The following theorem shows that if we drop the assumption of linear growth, then we do not have a finite energy solution to the time fractional stochastic heat equation (1.12). This theorem extends the result of Foondun and Parshad [14]

**Theorem 3.** *Suppose  $\inf_{z \in \mathbb{R}^d} u_0(z) > 0$  and  $\inf_{y \in \mathbb{R}^d} |\sigma(y)|/|y|^{1+\epsilon} > 0$ . Then, there is no finite-energy solution to the time fractional stochastic heat type equation (1.12).*

*Proof.* We will adapt the methods in [14] with many crucial changes. Let  $c := \inf_{y \in \mathbb{R}^d} |\sigma(y)|/|y|^{1+\epsilon}$  and  $l := \inf_{z \in \mathbb{R}^d} u_0(z)$ .

$$\begin{aligned} \mathbb{E}(|u_t(x)|^2) &= |(G_t * u_0)(x)|^2 + \int_0^t ds \int_{\mathbb{R}^d} dy [G_{t-s}(y-x)]^2 \mathbb{E}(\sigma^2(u_s(y))) \\ &\geq l^2 + c^2 \int_0^t \int_{\mathbb{R}^d} [G_{t-s}(y-x)]^2 \mathbb{E}(|u_s(y)|^{2(1+\epsilon)}) dy \end{aligned} \quad (5.1)$$

$$\geq l^2 + c^2 \int_0^t \int_{\mathbb{R}^d} [G_{t-s}(y-x)]^2 \inf_{y \in \mathbb{R}^d} [\mathbb{E}(|u_s(y)|^2)]^{1+\epsilon} dy, \quad (5.2)$$

with an application of the Jensen's inequality in the last inequality. If we let

$$I(t) := \inf_{y \in \mathbb{R}^d} \mathbb{E}(|u_s(t)(y)|^2) \quad (t \geq 0),$$

it satisfies the following inequality

$$\begin{aligned} I(t) &\geq l^2 + c^2 \int_0^t \|G_{t-s}\|_{L^2(\mathbb{R}^d)}^2 [I(s)]^{1+\epsilon} ds \\ &= l^2 + c^2 \int_0^t (t-s)^{-d\beta/\alpha} [I(s)]^{1+\epsilon} ds. \end{aligned} \quad (5.3)$$

Now we define the Laplace transform of  $I$  as

$$\tilde{I}(\theta) := \int_0^\infty e^{-\theta s} I(s) ds.$$

Because

$$\int_0^\infty e^{-\theta s} \|G_s\|_{L^2(\mathbb{R}^d)}^2 ds = \int_0^\infty e^{-\theta s} C^* s^{-\beta d/\alpha} ds = C_1 \theta^{-(1-\beta d/\alpha)}$$

for all  $\theta > 0$ , where  $C_1 = C^* \Gamma(1 - \beta d/\alpha)$ . It follows that

$$\tilde{I}(\theta) \geq \frac{l^2}{\theta} + c^2 C_1 \theta^{-(1-\beta d/\alpha)} \int_0^\infty e^{-\theta s} [I(s)]^{1+\epsilon} ds. \quad (5.4)$$

We multiply both sides by  $\theta$  and use Jensen's inequality to find that

$$\begin{aligned} \theta \tilde{I}(\theta) &\geq l^2 + c^2 C_1 \theta^{-(1-\beta d/\alpha)} \left[ \int_0^\infty \theta e^{-\theta s} [I(s)] ds \right]^{1+\epsilon} \\ &= l^2 + c^2 C_1 \theta^{-(1-\beta d/\alpha)} [\theta \tilde{I}(\theta)]^{1+\epsilon}, \end{aligned}$$

for all  $\theta > 0$ . It follows that  $\theta \tilde{I}(\theta) \geq l^2 > 0$ , and hence  $\tilde{I}(\theta) > 0$  for all  $\theta > 0$ . Moreover,  $[\theta \tilde{I}(\theta)]^{1+\epsilon} \geq l^{2\epsilon} \theta \tilde{I}(\theta)$ , and hence

$$\theta \tilde{I}(\theta) \geq l^2 + c^2 C_1 \theta^{-(1-\beta d/\alpha)} l^{2\epsilon} \theta \tilde{I}(\theta).$$

We shall show that

$$\tilde{I}(\theta) = \infty \quad \text{for } 0 < \theta \leq \theta_0 := (c^2 C_1 l^{2\epsilon})^{\alpha/(\alpha-\beta d)}.$$

Under the assumption that  $0 < \theta \leq \theta_0$ , the constant  $A := c^2 C_1 \theta^{-(1-\beta d/\alpha)} l^{2\epsilon}$  is greater than or equal to 1. With recursive argument we can show that for any positive integer  $n$ ,

$$\theta \tilde{I}(\theta) \geq l^2 + A^n \theta \tilde{I}(\theta).$$

Since  $A \geq 1, l > 0$  and  $n$  is arbitrary. We see that  $\theta \tilde{I}(\theta) = \infty$ . Which shows that  $\tilde{I}(\theta) = \infty$  for a given range of  $\theta$ .

Now we will show that there exists  $\theta_1 > \theta_0$  such that  $\tilde{I}(\theta_1) = \infty$ . Recall that  $\tilde{I}(\theta_0) = \infty$ . That means  $I(t) \geq ce^{\theta_0 t}$  for large  $t$ .

For  $t$  large enough, we have

$$\begin{aligned} I(t) &\geq l^2 + c^2 \int_0^t (t-s)^{-d\beta/\alpha} [I(s)]^{1+\epsilon} ds \\ &\geq l^2 + c^2 t^{-d\beta/\alpha} \int_0^t [I(s)]^{1+\epsilon} ds \end{aligned}$$

$$\begin{aligned} &\geq l^2 + c^2 t^{-d\beta/\alpha} \int_{t/2}^t e^{\theta_0(1+\epsilon)s} ds \\ &\geq l^2 + c^2 t^{-d\beta/\alpha} e^{\theta_0(1+\epsilon)t} \left(1 - e^{-\theta_0(1+\epsilon)t/2}\right). \end{aligned} \quad (5.5)$$

Using the inequality  $x/(1+x) < (1 - e^{-x})$  for  $x > -1$  we get

$$I(t) \geq l^2 + c^2 t^{-d\beta/\alpha} e^{\theta_0(1+\epsilon)t},$$

again for large enough  $t$ . Now if we take  $\theta_1 = \theta_0(1 + \epsilon)$ , we have  $\tilde{I}(\theta_1) = \infty$ . Since we can repeat this process over and over again there is no minimum  $\theta < \infty$  such that  $\tilde{I}(\theta) = \infty$ .

Now, if we assume there is a finite energy solution, we certainly have  $\tilde{I}(\theta_*) < \infty$  for some  $\theta_* > 0$ . With simple algebra we can show that

$$\tilde{I}(\theta) = \int_0^\infty e^{-\theta s} I(s) ds = \int_0^\infty e^{-(\theta - \theta_* + \theta_*)s} I(s) ds \leq \int_0^\infty e^{-\theta_* s} I(s) ds < \infty,$$

for  $\theta \geq \theta_*$ . That means  $\tilde{I}(\theta) < \infty$  for all  $\theta \geq \theta_*$ . But this contradicts the above argument. Hence there is no finite energy solution. This completes the proof.  $\square$

## 6. EQUIVALENCE OF TIME FRACTIONAL SPDES TO HIGHER ORDER SPDE'S

In this section we will give a connection of time fractional SPDE's and Higher order parabolic SPDE's.

Allouba [1; 2] considered the following SPDE( $k = 1, 2, \dots$ ):

$$\begin{aligned} \partial_t u_t^k(x) &= \sum_{j=1}^{2^k-1} \frac{t^{j/2^k-1}}{\Gamma(j/2^k)} \Delta_x^j u_0^k(x) + \Delta_x^{2^k} u_t^k(x) + \sigma(u^k) \dot{W}(t, x); \\ u_t^k(x)|_{t=0} &= u_0^k(x) \end{aligned} \quad (6.1)$$

Where  $\sigma$  satisfy Lipschitz and linear growth conditions. Allouba [2] showed that there exists a path wise unique strong (mild) solution in all space dimensions  $d = 1, 2, 3$  given by

$$\begin{aligned} u_t^k(x) &= \int_{\mathbb{R}} K_t^{BM^d, E^\beta}(x-y) u_0^k(y) dy \\ &+ \int_0^t \int_{\mathbb{R}} K_{t-r}^{BM^d, E^\beta}(x-y) \sigma(u_s^k(y)) W(dy dr) \end{aligned} \quad (6.2)$$

where  $K_t^{BM^d, E^\beta}(x-y)$  is the transition density of a  $d$ -dimensional process  $B(E^\beta)$ , here  $B$  is a  $d$ -dimensional Brownian motion, and  $E^\beta$  is inverse of a stable subordinator of index  $\beta = 2^{-k}$ :

$$K_t^{BM^d, E^\beta}(x-y) = \int_0^\infty \frac{e^{-\frac{|x-y|^2}{4s}}}{\sqrt{4\pi s}} f_{E_t}(s) ds.$$

Allouba [1; 2] also showed that the Hölder continuity exponent (time, space) of the mild solution of (6.1) is

$$\left( \left( \frac{2\beta^{-1} - d}{4\beta^{-1}} \right)^-, \left( \frac{4-d}{2} \wedge 1 \right)^- \right),$$

(where  $\beta = 2^{-k}$ ) in space dimensions  $d = 1, 2, 3$ .

Since Baeumer et al. [5] showed that  $G_t(x - y) = K_t^{BM^d, E^\beta}(x - y)$  for  $\alpha = 2$  and

$$K_t^{Y, E^\beta}(x - y) = \int_0^\infty p_{Y(s)}(x - y) f_{E_t}(s) ds = G_t(x - y), \quad \text{for } 0 < \alpha < 2,$$

we obtain the next result

**Theorem 4.** *Let  $L_x = -\nu(-\Delta)^{\alpha/2}$  for  $\alpha \in (0, 2]$  and suppose  $\sigma$  is Lipschitz. For any  $k = 1, 2, 3, \dots$  both the higher order SPDE*

$$\partial_t u_t^k(x) = \sum_{j=1}^{2^k-1} \frac{t^{j/2^k-1}}{\Gamma(j/2^k)} L_x^j u_0^k(x) + L_x^{2^k} u_t^k(x) + \sigma(u_k) \dot{W}(t, x); \quad (6.3)$$

$$u_t^k(x)|_{t=0} = u_0^k(x),$$

and the time fractional SPDE

$$\partial_t^{1/2^k} u_t^k(x) = L_x u_t^k(x) + I_t^{1-1/2^k} [\sigma(u^k) \dot{W}(t, x)]; \quad u_t^k(x)|_{t=0} = u_0^k(x), \quad (6.4)$$

have the same unique mild solution given by

$$\begin{aligned} u_t^k(x) &= \int_{\mathbb{R}} K_t^{Y, E^\beta}(x - y) u_0^k(y) dy \\ &+ \int_0^t \int_{\mathbb{R}} K_{t-r}^{Y, E^\beta}(x - y) \sigma(u_s^k(y)) W(dy dr). \end{aligned} \quad (6.5)$$

**Remark 6.** *Theorem 4 extends the particular case considered by Baeumer et al. [5]—the case of  $\sigma \equiv 0$ —to the equivalence of SPDE's. Allouba [2, equation (1.8)] mentions equivalence of (6.1) to the equation*

$$\partial_t^{1/2^k} u_t^k(x) = \Delta_x u_t^k(x) + \sigma(u^k) \dot{W}(t, x); \quad u_t^k(x)|_{t=0} = u_0^k(x). \quad (6.6)$$

Our theorem 4 gives the correct equivalence.

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DEPARTMENT OF MATHEMATICS, GEORGIA COLLEGE & STATE UNIVERSITY, MILLEDGEVILLE, GA 31061

*E-mail address:* jebessa.mijena@gcsu.edu

DEPARTMENT OF MATHEMATICS AND STATISTICS, AUBURN UNIVERSITY, AUBURN, AL 36849, USA

*E-mail address:* nane@auburn.edu